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Two-Parameter Ridge Estimation in Seemingly Unrelated Regression Models

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Abstract:

Seemingly Unrelated Regression (SUR) models are used when several linear regression equations are investigated simultaneously. To reduce the impact of multicollinearity in SUR models, the one-parameter ridge (Ridge-1) estimator has been proposed. As a generalization, this paper introduces the two-parameter ridge (Ridge-2) estimator for SUR models with multicollinearity. The performance of the proposed estimator is compared with Generalized Least Squares (GLS) and Ridge-1 via simulations. A real dataset on chronic renal failure is used for illustration. The results show that the Ridge-2 estimator outperforms both GLS and Ridge-1 under severe multicollinearity.

Keywords: Multicollinearity, One-parameter ridge, Seemingly unrelated regression, Two-parameter ridge, Mean squared error.

Mathematics Subject Classification (2010): 62J07, 62J05.

1 Introduction

Multicollinearity among explanatory variables is a well-known problem in linear regression that inflates variances and distorts coefficient estimates. Ridge regression, introduced by [Hoerl and Kennard \(1970\)](#),

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addresses this issue by adding a small constant $k > 0$ to the diagonal of the $X'X$ matrix. While effective, the one-parameter ridge (Ridge-1) estimator often suffers from poor model fit. To overcome this limitation, [Lipovetsky and Conklin \(2005\)](#) proposed the two-parameter ridge (Ridge-2) estimator by introducing a second tuning parameter that maximizes the coefficient of determination. [Toker and Kaçiranlar \(2013\)](#) derived mean squared error (MSE) superiority conditions for Ridge-2 over both Ridge-1 and generalized least squares (GLS), demonstrating its advantages in ordinary linear models.

In many applied settings, multiple regression equations share correlated error structures, making seemingly unrelated regression (SUR) models, pioneered by [Zellner \(1962\)](#), more appropriate. Multicollinearity in SUR models is particularly problematic, as it degrades the efficiency of GLS estimators. [Firinguetti \(1997\)](#) adapted ridge regression to SUR systems, with asymptotic properties established by [Firinguetti and Rubio \(2008\)](#). [Alkhamisi and Shukur \(2008\)](#) further developed ridge parameter selection methods specifically for SUR models.

To the best of our knowledge, Ridge-2 estimation has not been extended to SUR models with multicollinearity. This paper fills this gap by proposing a two-parameter ridge estimator for SUR systems and deriving conditions under which it outperforms both GLS and Ridge-1 in terms of MSE. Extensive simulations and a real-data application on chronic renal failure demonstrate the superiority of the proposed estimator under severe multicollinearity.

2 The SUR models and their solutions

Consider the system of g equations $Y_i = X_i\beta_i + e_i$ with $E(e_i) = 0$ and $E(e_ie_j') = \sigma_{ij}I_n$ is compactly written as

$$Y = X\beta + e, \quad (2.1)$$

where X is block diagonal and $E(ee') = \Sigma \otimes I_n$. Using the transformation $Y^* = (\Sigma^{-1/2} \otimes I_n)Y$, $X^* = (\Sigma^{-1/2} \otimes I_n)X$, the model becomes the standardized SUR model

$$Y^* = X^*\beta + e^*, \quad (2.2)$$

with uncorrelated, unit-variance errors. The GLS estimator is

$$\hat{\beta}_{\text{GLS}}^{\text{SUR}} = D^{-1}c, \quad (2.3)$$

where $D = X^{*'}X^*$ and $c = X^{*'}Y^*$. The Ridge-1 estimator is

$$\hat{\beta}_{\text{R1}}^{\text{SUR}}(k) = (D + kI_p)^{-1}c, \quad k > 0, \quad (2.4)$$

and the corresponding multiple determination coefficient is

$$\hat{R}_{SUR}^2(k) = 2c'(D + kI_p)^{-1}c - c'(D + kI_p)^{-1}D(D + kI_p)^{-1}c, \quad (2.5)$$

both reducing to the GLS forms when $k = 0$.

2.1 Proposed Ridge-2 Estimator

The Ridge-2 estimator is presented as:

$$\hat{\beta}_{R2}^{SUR}(k, q) = q(D + kI_p)^{-1}c \quad (2.6)$$

where q is a constant parameter. The coefficient of multiple determination corresponding to the Ridge-2 estimator (2.6), as a function of (k, q) , is obtained as

$$\hat{R}_{SUR}^2(k, q) = 2qc'(D + kI_p)^{-1}c - q^2c'(D + kI_p)^{-1}D(D + kI_p)^{-1}c. \quad (2.7)$$

For fixed k , maximizing $\hat{R}_{SUR}^2(k, q)$ with respect to q yields the optimal second parameter

$$q_{\max}^{SUR} = \frac{c'(D + kI_p)^{-1}c}{c'(D + kI_p)^{-1}D(D + kI_p)^{-1}c}. \quad (2.8)$$

Now, setting $q = q_{\max}^{SUR}$ in (2.6) and (2.7) yields

$$\hat{\beta}_{R2}^{SUR}(k) := \hat{\beta}_{R2}^{SUR}(k, q_{\max}^{SUR}) = \frac{c'(D + kI_p)^{-1}c}{c'(D + kI_p)^{-1}D(D + kI_p)^{-1}c}(D + kI_p)^{-1}c \quad (2.9)$$

and

$$\hat{R}_{SUR}^2(k, q_{\max}^{SUR}) = \frac{[c'(D + kI_p)^{-1}c]^2}{c'(D + kI_p)^{-1}D(D + kI_p)^{-1}c}, \quad (2.10)$$

respectively. Since $D > 0$, one can show,

$$\hat{R}_{SUR}^2(k, q_{\max}^{SUR}) - \hat{R}_{SUR}^2(k) = \frac{[c'(D + kI_p)^{-1}c - c'(D + kI_p)^{-1}D(D + kI_p)^{-1}c]^2}{c'(D + kI_p)^{-1}D(D + kI_p)^{-1}c} \geq 0.$$

Thus, for any fixed k , the Ridge-2 model always achieves a higher coefficient of determination than Ridge-1.

2.2 Unknown covariance case

In practice, Σ is unknown and must be estimated. Following [Firingueti \(1997\)](#), we estimate its elements by

$$\hat{\sigma}_{ij} = \frac{Y_i'(I_n - X_i(X_i'X_i)^{-1}X_i')(I_n - X_j(X_j'X_j)^{-1}X_j')Y_j}{n}, \quad i, j = 1, 2, \dots, g \quad (2.11)$$

Replacing Σ with $\hat{\Sigma}$ yields the operational GLS (OGLS), Ridge-1 (OR1), and Ridge-2 (OR2) estimators:

$$\hat{\beta}_{OGLS}^{SUR} = \hat{D}^{-1}\hat{c}, \quad (2.12)$$

$$\hat{\beta}_{OR1}^{SUR}(k) = (\hat{D} + kI_p)^{-1}\hat{c}, \quad (2.13)$$

$$\hat{\beta}_{OR2}^{SUR}(k) = \hat{q}_{\max}^{SUR}(\hat{D} + kI_p)^{-1}\hat{c}, \quad (2.14)$$

where $\hat{D} = X'(\hat{\Sigma}^{-1} \otimes I_n)X$, $\hat{c} = X'(\hat{\Sigma}^{-1} \otimes I_n)Y$, and $\hat{q}_{max}^{SUR} = \frac{\hat{c}'(\hat{D} + kI_p)^{-1}\hat{c}}{\hat{c}'(\hat{D} + kI_p)^{-1}\hat{D}(\hat{D} + kI_p)^{-1}\hat{c}}$.

2.3 Choosing k

We adopt the ridge parameters for Ridge-1 and Ridge-2 from [Alkhamisi and Shukur \(2008\)](#). The proposed values are functions of $\hat{a} = (Z'Z)^{-1}Z'Y^*$, where $Z = X^*T$ and T is the eigenvector matrix of D :

$$\begin{aligned} k_{SHK} &= [\max \text{diag}(\hat{a})^2]^{-1}, \quad k_{Smed} = [\text{median} \text{diag}(\hat{a})^{-2}], \\ k_{Sqarith} &= [\text{mean} \text{diag}(\hat{a})^{-\frac{1}{2}}], \quad k_{Sqmax} = [\max \text{diag}(\hat{a})^{-\frac{1}{2}}], \quad k_{Smax} = [\max \text{diag}(\hat{a})^{-2}]. \end{aligned} \quad (2.15)$$

When Σ is unknown, these quantities are estimated by substituting $\hat{\Sigma}$.

3 Investigation of the solutions for the SUR models

We compare Ridge-2 with GLS and Ridge-1 via matrix mean squared error, $MSE(\hat{\beta}) = \text{Var}(\hat{\beta}) + \text{Bias}(\hat{\beta})\text{Bias}(\hat{\beta})'$.

Using the spectral decomposition $D = TKT'$ with $K = \text{diag}(\kappa_1, \dots, \kappa_p)$, the estimators become $\hat{\beta}_{GLS} = TK^{-1}T'c$, $\hat{\beta}_{R1}(k) = (TKT' + kI_p)^{-1}c$, and $\hat{\beta}_{R2}(k, q) = q(TKT' + kI_p)^{-1}c$. Their MSE matrices are [Toker and Kaçiranlar \(2013\)](#):

$$MSE(\hat{\beta}_{GLS}) = TK^{-1}T', \quad (3.1)$$

$$MSE(\hat{\beta}_{R1}(k)) = V_k + b_2b_2', \quad (3.2)$$

$$MSE(\hat{\beta}_{R2}(k, q)) = q^2V_k + b_1b_1', \quad (3.3)$$

where $V_k = (TKT' + kI_p)^{-1}TKT'(TKT' + kI_p)^{-1}$, $b_1 = [q(TKT' + kI_p)^{-1}TKT' - I_p]\beta$, and $b_2 = -k(TKT' + kI_p)^{-1}\beta$. The derivations mirror those for ordinary linear models; we state the key results.

Lemma 3.1. *Let $k > (q-1)\max_j(\kappa_j)$ and $b_1 = \text{Bias}(\hat{\beta}_{R2}(k, q))$. Then $MSE(\hat{\beta}_{GLS}) - MSE(\hat{\beta}_{R2}(k, q)) > 0$ iff $b_1'(D^{-1} - q^2V_k)^{-1}b_1 < 1$.*

Lemma 3.2. *Let $b_1 = \text{Bias}(\hat{\beta}_{R2}(k, q))$ and $b_2 = \text{Bias}(\hat{\beta}_{R1}(k))$. If $b_2'V_k^{-1}b_2 < (q^2-1)$, then $MSE(\hat{\beta}_{R2}(k, q)) - MSE(\hat{\beta}_{R1}(k)) > 0$.*

Lemma 3.3. *Let $b_1 = \text{Bias}(\hat{\beta}_{R2}(k, q))$ and $b_2 = \text{Bias}(\hat{\beta}_{R1}(k))$. Then $MSE(\hat{\beta}_{R2}(k, q)) - MSE(\hat{\beta}_{R1}(k)) > 0$ iff $b_2'((q^2-1)V_k + b_1b_1')b_1 < 1$.*

4 Simulation study

A simulation study compares GLS, Ridge-1, and Ridge-2 estimators in SUR models via total MSE: $TMSE(\hat{\beta}) = \text{trace } E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)']$. The simulation parameters used in this study are given in Table 1.

Table 1: Simulation parameters.

Simulation factor	Symbol	Values
Number of observations	n	10, 50, 100
Number of equations	g	2, 3
Number of explanatory variables	p_0	10, 50, 100
The model parameters	β	$\mathbf{1}_{p_0 g}, (\mathbf{1}'_{\frac{p_0 g}{2}}, \mathbf{2}'_{\frac{p_0 g}{2}})'$
Correlation among the explanatory variables	q_x	0.5, 0.98
Correlation among the response variables	q_y	0.3, 0.9
Ridge parameter	k	$k_{\text{SHK}}, k_{\text{Skmed}}, k_{\text{Sqarith}}, k_{\text{Sqmax}}, k_{\text{Smax}}$

Design matrices are $X_i = [\mathbf{1}_n \mid X_i^*]$ with rows of X_i^* drawn from $N_{p_0-1}(\mathbf{0}, \Sigma_X)$, $\Sigma_X = \rho_x J_{p_0-1} + (1 - \rho_x)I_{p_0-1}$. Error covariance is $\Sigma = \rho_y J_g + (1 - \rho_y)I_g$, with $e \sim N_{ng}(\mathbf{0}, \Sigma \otimes I_n)$. Each configuration ran 10,000 times per Algorithm 1.

Algorithm 1. *Computing the TMSE of GLS, Ridge-1 and Ridge-2 estimators:*

- *Step 1: Generate $X_1, X_2, \dots, X_g$ as in the design matrices.*
- *Step 2: Generate $e = (e'_1, \dots, e'_g)'$ from $N_{ng}(\mathbf{0}_{ng}, \Sigma \otimes I_n)$.*
- *Step 3: Calculate $Y_1, Y_2, \dots, Y_g$ using $Y_i = X_i \beta_i + e_i$, $i = 1, 2, \dots, g$.*
- *Step 4: Calculate $\hat{\beta}_{\text{OGLS}}^{\text{SUR}(r)}$, $\hat{\beta}_{\text{OR1}}^{\text{SUR}(r)}(k)$ and $\hat{\beta}_{\text{OR2}}^{\text{SUR}(r)}(k)$ using (2.12), (2.13) and (2.14), respectively.*
- *Step 5: Repeat the steps 1–4 for $r = 1, 2, \dots, 10000$.*
- *Step 6: Calculate the empirical TMSE (ETMSE) of the GLS, Ridge-1 and Ridge-2 estimators as*

$$ETMSE(\hat{\beta}_{\text{OGLS}}^{\text{SUR}}) = \frac{1}{10000} \sum_{r=1}^{10000} (\hat{\beta}_{\text{OGLS}}^{\text{SUR}(r)} - \beta)' (\hat{\beta}_{\text{OGLS}}^{\text{SUR}(r)} - \beta), \quad (4.1)$$

$$ETMSE(\hat{\beta}_{\text{OR1}}^{\text{SUR}}(k)) = \frac{1}{10000} \sum_{r=1}^{10000} (\hat{\beta}_{\text{OR1}}^{\text{SUR}(r)}(k) - \beta)' (\hat{\beta}_{\text{OR1}}^{\text{SUR}(r)}(k) - \beta), \quad (4.2)$$

$$ETMSE(\hat{\beta}_{\text{OR2}}^{\text{SUR}}(k)) = \frac{1}{10000} \sum_{r=1}^{10000} (\hat{\beta}_{\text{OR2}}^{\text{SUR}(r)}(k) - \beta)' (\hat{\beta}_{\text{OR2}}^{\text{SUR}(r)}(k) - \beta), \quad (4.3)$$

respectively.

For all parameter combinations, the ratios $R2R1 := \frac{ETMSE(\hat{\beta}_{OR2}^{SUR}(k))}{ETMSE(\hat{\beta}_{OR1}^{SUR}(k))}$ and $R2GLS := \frac{ETMSE(\hat{\beta}_{OR2}^{SUR}(k))}{ETMSE(\hat{\beta}_{OGLS}^{SUR}(k))}$ were computed for SUR models with $g = 2$ and $g = 3$ equations, each with β equal to 1_{p_0g} and $(1'_{\frac{p_0g}{2}}, 2'_{\frac{p_0g}{2}})'$.

Ridge-2 vs. Ridge-1: Ridge-2 with k_{Smax} , $k_{Sqarith}$ and k_{SHK} uniformly outperforms Ridge-1 across all configurations. Ridge-1 is marginally better only when $\rho_x = 0.5$ with small p_0 using k_{Smed} or k_{Sqmax} . As ρ_x increases to 0.98, Ridge-2's superiority becomes pronounced.

Ridge-2 vs. GLS: Ridge-2 with k_{Sqmax} slightly surpasses GLS. With k_{SHK} or k_{Smed} , both perform comparably. Under $k_{Sqarith}$ or k_{Smax} , Ridge-2 excels in severe multicollinearity but underperforms when multicollinearity is mild. Notably, Ridge-2 dramatically outperforms GLS when p_0 approaches n .

Effect of ρ_y : Ridge-2 performs best under weak cross-equation error correlation ($\rho_y = 0.3$). In summary, Ridge-2 is superior or equivalent to both competitors under high multicollinearity and when $p_0 \approx n$, across all values of ρ_x , ρ_y , and k .

5 Real Data Application

We analyze data from a medical study at Tabriz University of Medical Sciences on the effect of chronic renal failure (CRF) on visual evoked potentials (VEP) in 30 peritoneal dialysis patients. Since optic nerve involvement in CRF is mainly subclinical, VEP is used to evaluate it [Fraser and Arieff \(1988\)](#); [Burn and Bates \(1998\)](#); [Brouns and De Deyn \(2004\)](#); [Lacerda et al. \(2010\)](#). The VEP test assesses optic nerve function via P100 latency (P100lat) and P100 amplitude (P100amp), as renal failure causes accumulation of toxic substances (urea, creatinine, potassium) normally excreted by kidneys, leading to subclinical optic nerve damage detectable only by VEP.

Five laboratory predictors ($p = 6$ with intercept) are included: urea (Ur), creatinine (Cr), potassium (K), hemoglobin (Hb), and hematocrit (Hct). Multicollinearity among these variables makes ordinary least squares problematic. We assess it via variance inflation factor (VIF), defined for variable X_j as $VIF_j = (1 - R_j^2)^{-1}$, where R_j^2 is obtained by regressing X_j on all other predictors, with VIFs exceeding 10 indicating serious multicollinearity.

Table 2: The values of VIF for explanatory variables.

Factor	Hct	Cr	Hb	K	Ur
VIF	18.166	2.232	20.095	1.177	1.776

Table 2 confirms severe multicollinearity ($VIF > 10$ for Hct and Hb; correlation = 0.97), causing GLS to yield opposite signs (Table 3) that contradict clinical expectation. We therefore discard GLS in favor of

Ridge-1 and Ridge-2.

Based on the simulation study, Ridge-2 with $k = k_{\text{Smax}}$ performed best. We therefore apply GLS, Ridge-1 with k_{Smax} (SmaxR1), and Ridge-2 with k_{Smax} (SmaxR2).

Table 3: The values of regression coefficients for laboratory factors.

	P100lat			P100amp		
	GLS	SmaxR1	SmaxR2	GLS	SmaxR1	SmaxR2
Intercept	56.6838	0.0781	1.09126	0.1556	2.0792	0.6959
Ur	-0.2040	-0.1207	0.0780	2.0154	0.5298	0.4572
Cr	1.0911	0.0283	-0.1207	0.5988	0.3912	0.5369
K	0.0780	1.4002	0.0225	0.4632	0.0191	0.7310
Hb	-0.2444	1.2281	1.4032	0.1159	0.2542	0.0201
Hct	0.0226	0.6924	2.0052	0.7697	0.0903	0.2676

For P100lat, SmaxR2 yields the largest coefficients for Hct (2.0052) and Hb (1.4032), consistent with clinical evidence linking high Hct to visual impairment, while other predictors show small effects. For P100amp, SmaxR2 coefficients indicate a positive association between biochemical factors and P100 amplitude. The SmaxR2 estimator thus provides clinically interpretable results and resolves the sign reversal observed under GLS.

Discussion and Conclusion

Multicollinearity in SUR models distorts GLS estimates, reversing signs and inflating standard errors. We propose a two-parameter ridge (Ridge-2) estimator, deriving conditions for its superiority over GLS and Ridge-1. Simulations confirm that Ridge-2 with $k = k_{\text{Smax}}$ performs best under severe multicollinearity, making it the recommended choice for SUR estimation with highly correlated predictors.

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